```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore
data={
    "student": ['A','B','C','D','E','F','G','H','I','J','K','L','M','N','O'],
    "study_hours":[2,3,4,5,4,6,5,3,4,5,6,4,5,30,3]
}
df=pd.DataFrame(data)
print("Original Data set")
display(df)
```

```
df["study_hours"].describe()
```

```
plt.figure(figsize=(8,4))
sns.boxplot(X=df["study_hours"])
plt.title("study_hours-Boxplot")
plt.xlabel("study hours")
plt.show()
```

```
Q1=df["study_hours"].quantile(0.25)
Q3=df["study_hours"].quantile(0.75)
print("Q1:",Q1)
print("Q3:",Q3)
```

```
Q1: 3.5
Q3: 5.0
```

```
IQR=Q3-Q1
print("IQR:",IQR)
```

```
IQR: 1.5
```

```
lower_bound=Q1-1.5*IQR
upper_bound=Q3+1.5*IQR
print("Lower Bound:",lower_bound)
print("Upper Bound:",upper_bound)
```

```
Lower Bound: 1.25
Upper Bound: 7.25
```

```
outliers_iqr=df[
    (df["study_hours"]<lower_bound) | (df["study_hours"]>upper_bound)
]
print("Outliers using IQR:")
display(outliers_iqr)
```

```
normal_data_iqr=df[
    (df["study_hours"]>=lower_bound) & (df["study_hours"]<=upper_bound)
]
print("Normal Data using IQR:")
display(normal_data_iqr)
```

```
df["Z_Score"]=zscore(df[("study_hours")])
print("Data with Z-score")
display(df)
```

```
z_outliers=df[
    (df["Z_Score"]<-3) | (df["Z_Score"]>3)
]
print("Outliers using Z-score:")
display(z_outliers)
```

```
print("IQR Outliers:")
display(outliers_iqr)
```

```
print("Z-score Outliers:")
display(z_outliers)
print("Normal Data using IQR:")
display(normal_data_iqr)
```

```
df_capped=df.copy()
df_capped["study_hours"]=df_capped["study_hours"].clip(
    lower=lower_bound,upper=upper_bound
)
df_capped
```

```
df_without_outliers=df[(df["Z_Score"]>=-3) & (df["Z_score"]<=3)]
print("Data without Outliers:")
display(df_without_outliers)
```

Start coding or generate with AI.